

LD10 Variable and Method Flow

How data, variables and models move through the notebook

1. Global project variables

Variable	Meaning	Used for
FAST_MODE	Controls whether the notebook runs quickly or fully	Changes tuning size and Optuna trials
DATA_DIR	Main data folder	Stores Bank Marketing and House Prices data
RANDOM_STATE	Fixed random seed	Keeps splits and model results reproducible
N_OPTUNA_TRIALS	Number of Optuna trials	10 in this executed run

2. Bank Marketing classification flow

Step	Key variables	Purpose
Load data	bank_data	Raw Bank Marketing dataset with customer and campaign fields
Find target	bank_target_column = deposit	Defines what the model predicts
Separate inputs and target	X_bank, y_bank_raw	Splits features from the label
Encode categories	label_encoders, X_bank	Turns text categories into numeric values
Encode target	target_encoder, y_bank	Maps yes/no to numeric labels
Train/test split	X_bank_train, X_bank_test	Creates fair unseen test data
Train models	xgb_classifier, rf_classifier	Compares boosted trees with Random Forest
Evaluate	classification_results	Stores Accuracy and F1-score
Feature importance	top_xgb_importance	Identifies strongest drivers

3. House Prices regression flow

Step	Key variables	Purpose
Load data	house_data	Raw House Prices dataset
Set target	y_house = SalePrice	Defines the price to predict
Split features	X_house	Property attributes used as inputs
Identify column types	numeric_house_columns, categorical_house_columns	Separates numeric and categorical preprocessing
Train/validation split	X_house_train, X_house_valid	Creates validation set for RMSE
Preprocess for XGBoost/LightGBM	house_preprocessor	Imputes missing values and one-hot encodes categories
XGBoost baseline	xgb_regressor_baseline	Creates first regression benchmark
GridSearchCV	xgb_grid_search	Finds a tuned XGBoost setup
CatBoost	catboost_regressor	Uses raw categorical columns with cat_features
LightGBM + Optuna	lgbm_study, tuned_lgbm	Tunes LightGBM with smart search
Final ensemble	ensemble_pred	Averages XGBoost, CatBoost and LightGBM predictions
Final comparison	regression_results	Ranks models by RMSE

4. Metrics flow

Metric	Problem type	Why used
Accuracy	Classification	Shows overall correct predictions
F1-score	Classification	Balances precision and recall for campaign conversion
Feature importance	Classification	Explains which variables the model used most
RMSE	Regression	Measures prediction error in house-price units
Best iteration	Boosting training	Shows where early stopping found the best model